

# Companion to Bird & de Moor, Algebra of Programming

## §1. Programs

### §1.1. Datatypes

## §1.2. Natural numbers

## §1.3. Lists

## §1.4. Trees

## §1.5. Inverses

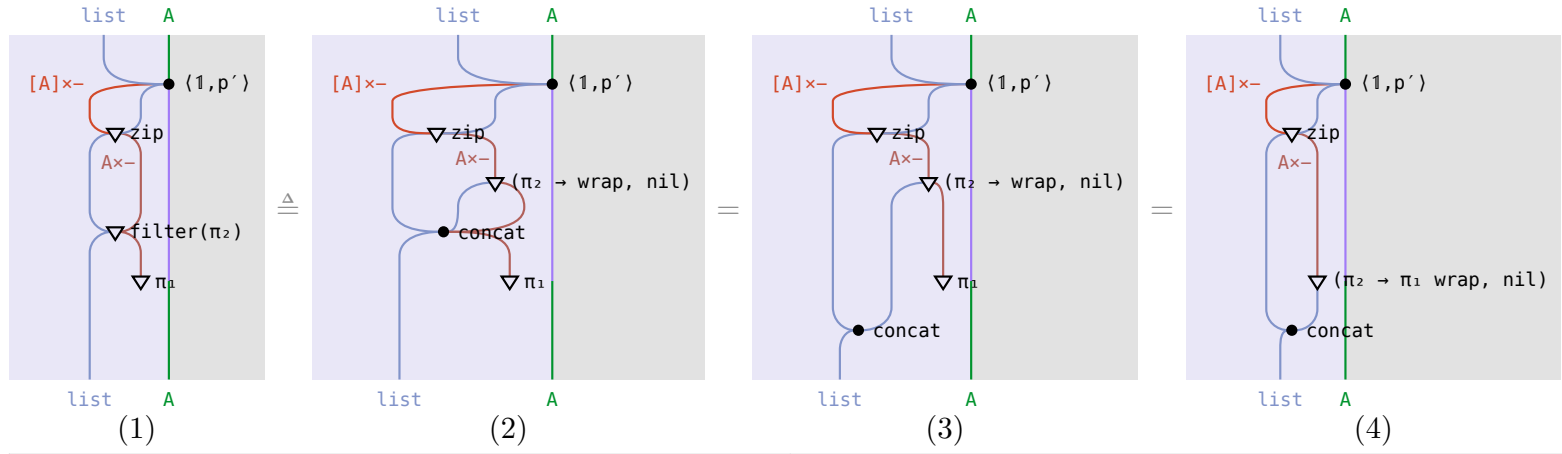
## §1.6. Polymorphic functions

## §1.7. Pointwise and point-free

$$\langle 1, p' \rangle \text{ zip filter}(\pi_2) \pi_1' = \text{filter}(p) \quad (1.7a)$$

pairing each element with its answer under  $p$  ( $\langle 1, p' \rangle \text{ zip}$ ), keeping the pairs answered **true** ( $\text{filter}(\pi_2)$ ) and dropping the answers ( $\pi_1'$ ) is  $\text{filter}(p)$

the starting composite



|            |                                                                                                                     |                                                                                                                                           |
|------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 1          | $\langle 1, p' \rangle \text{ zip filter}(\pi_2) \pi_1'$                                                            | definition of filter: $\text{filter}(p) \triangleq (p \rightarrow \text{wrap}, \text{nil})'$<br>concat                                    |
| 2          | $\triangleq \langle 1, p' \rangle \text{ zip } (\pi_2 \rightarrow \text{wrap}, \text{nil})' \text{ concat } \pi_1'$ | (1.6): $\text{concat } f' = f' \text{ concat}$<br>(1.8) backwards: $f' g' = (fg)'$                                                        |
| 3          | $= \langle 1, p' \rangle \text{ zip } (\pi_2 \rightarrow \text{wrap}, \text{nil}) \pi_1'' \text{ concat}$           | (1.11): $(p \rightarrow f, g)h = (p \rightarrow fh, gh)$ ; (1.5): $\text{wrap } f' = f$<br>wrap; (1.4): $\text{nil } f' = \text{nil}$     |
| 4          | $= \langle 1, p' \rangle \text{ zip } (\pi_2 \rightarrow \pi_1 \text{ wrap}, \text{nil})' \text{ concat}$           | (1.9) backwards: $1 = 1'$                                                                                                                 |
| <p>(5)</p> |                                                                                                                     |                                                                                                                                           |
| <p>(6)</p> |                                                                                                                     |                                                                                                                                           |
| <p>(7)</p> |                                                                                                                     |                                                                                                                                           |
| <p>(8)</p> |                                                                                                                     |                                                                                                                                           |
| 5          | $= \langle 1', p' \rangle \text{ zip } (\pi_2 \rightarrow \pi_1 \text{ wrap}, \text{nil})' \text{ concat}$          | (1.7): $\langle f', g' \rangle \text{ zip} = \langle f, g \rangle'$<br>(1.8) backwards: $f' g' = (fg)'$                                   |
| 6          | $= \langle 1, p \rangle (\pi_2 \rightarrow \pi_1 \text{ wrap}, \text{nil})' \text{ concat}$                         | (1.10): $h(p \rightarrow f, g) = (hp \rightarrow hf, hg)$ ; (1.1): $\langle f, g \rangle \pi_1 = f$ ; (1.3): $f \text{ nil} = \text{nil}$ |
| 7          | $= (p \rightarrow \text{wrap}, \text{nil})' \text{ concat}$                                                         | (1.12): $1 f = f$                                                                                                                         |
| 8          | $= \text{filter}(p)$                                                                                                | definition of filter                                                                                                                      |